

Git Feature-Branch Workflow — Lab Guide

Software Engineering (24PCA402) · Version Control

Objective

Practice a feature-branch workflow: branching, committing, merging, and resolving a conflict deliberately introduced for this exercise.

Steps

- Clone the starter repository and create a branch: `git checkout -b feature/login-form`
- Make small, focused commits with clear messages
- Push the branch and open a pull request against main
- Merge a second branch that touches the same file to create a conflict
- Resolve the conflict locally, then complete the merge

Key Commands

- `git branch`, `git checkout -b` — create and switch branches
- `git merge` — combine histories
- `git rebase -i` — rewrite commit history before sharing
- `git log --graph --oneline --all` — visualize branch topology

Deliverable

Submit the repository URL along with a short write-up (half a page) describing the conflict you hit and how you resolved it.